DockEye

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- One fixed camera. Recorded footage. Human review.
- Motion → model → tracking → events → review

Problem and goal

- Understand when people use the dock and when boats pass.
- Distinguish walking and standing from fishing activity.
- Produce reviewable events, timestamps and supporting snapshots.
- Detect activity; do not identify people or decide whether access is authorised.

Current release: what exists

- Validated zones, per-zone MOG2 and explicit shadow removal.
- Local YOLO adapter, motion gating and active-track reclassification.
- Track IDs, temporal event logic and directional boat counting.
- CSV, event snapshots, annotated video and operator controls.
- Custom fishing model accuracy remains unverified; rod candidates are experimental.

Architecture

- Recorded video → validated dock/water zones → motion regions.
- Motion + active track crops → YOLO → optional rod candidate geometry.
- One-to-one tracking → temporal evidence / boat-line crossing.
- Confirmed output → CSV + one snapshot → operator review.
- Active tracks are revisited even when motion stops.

Fishing: two different methods

- Learned method: custom person_fishing class, trained on reviewed labels.
- Alternative: detect a person and a long angled line near the upper body/hands.
- Require repeated evidence over time; reset on negatives or long evidence gaps.
- Rod output is fishing_candidate, not verified fishing.
- Hooks, bags, branches and railings are important negative examples.

Project scope

- In scope: one fixed camera and offline recordings of this dock.
- Detection, track continuity, line-crossing counts and event review.
- Optional measured dock-plane positions in metres; not body height.
- Daylight/dusk and wet/dry conditions are evaluation targets.
- Out of scope: live alerts, personal identity, night vision and multiple cameras.

Operator use cases

- UC1: define zones once and reload them for the same view.
- UC2: analyse a clip, inspect boxes/IDs and tune motion parameters.
- UC3: review time-confirmed fishing or explicitly tentative rod candidates.
- UC4: count boats crossing a chosen line and compare with a manual count.
- UC5: inspect CSV and matching snapshots; export an annotated video.

Data and training workflow

- Review labels: boat, person, person_fishing; include explicit negatives.
- Split by source recording/session, never neighbouring frames.
- Validate provenance and class coverage in train, validation and test.
- Fine-tune local pretrained YOLO weights; evaluate held-out clips.
- Public source footage is available; a reviewed three-class dataset is still needed.

Verification and remaining evidence

- Automated tests cover video → tracking → event → CSV/snapshot/export.
- Actual pretrained inference found people in an official reference image.
- Synthetic one-epoch training produced weights and per-class reports.
- Synthetic training metrics were zero; this is not a useful fishing model.
- Dock accuracy, event recall and empty-dock false-positive rate remain to be measured.

Risks and next work

- Walking-person recording: 3 false rod candidates from background lines.
- Track IDs can change; once-per-ID counting is not global re-identification.
- Validate rod candidates against hooks, bags, diagonal structures and no-rod clips.
- Collect independent labelled fishing recordings and tune on validation only.
- Measure per-class precision/recall, event matches and false events per hour.

Sources and demonstration

- Implementation, tests and data schemas: Herator/DockEye, current branch.
- Ultralytics YOLO11n: official pretrained person/boat starting model.
- Project recordings: shared Drive folder linked from the repository.
- Roboflow fishing recognition: candidate source; access/licence/weights unverified.
- Demo: explain the model/method, show outputs, and distinguish candidates from confirmed fishing.